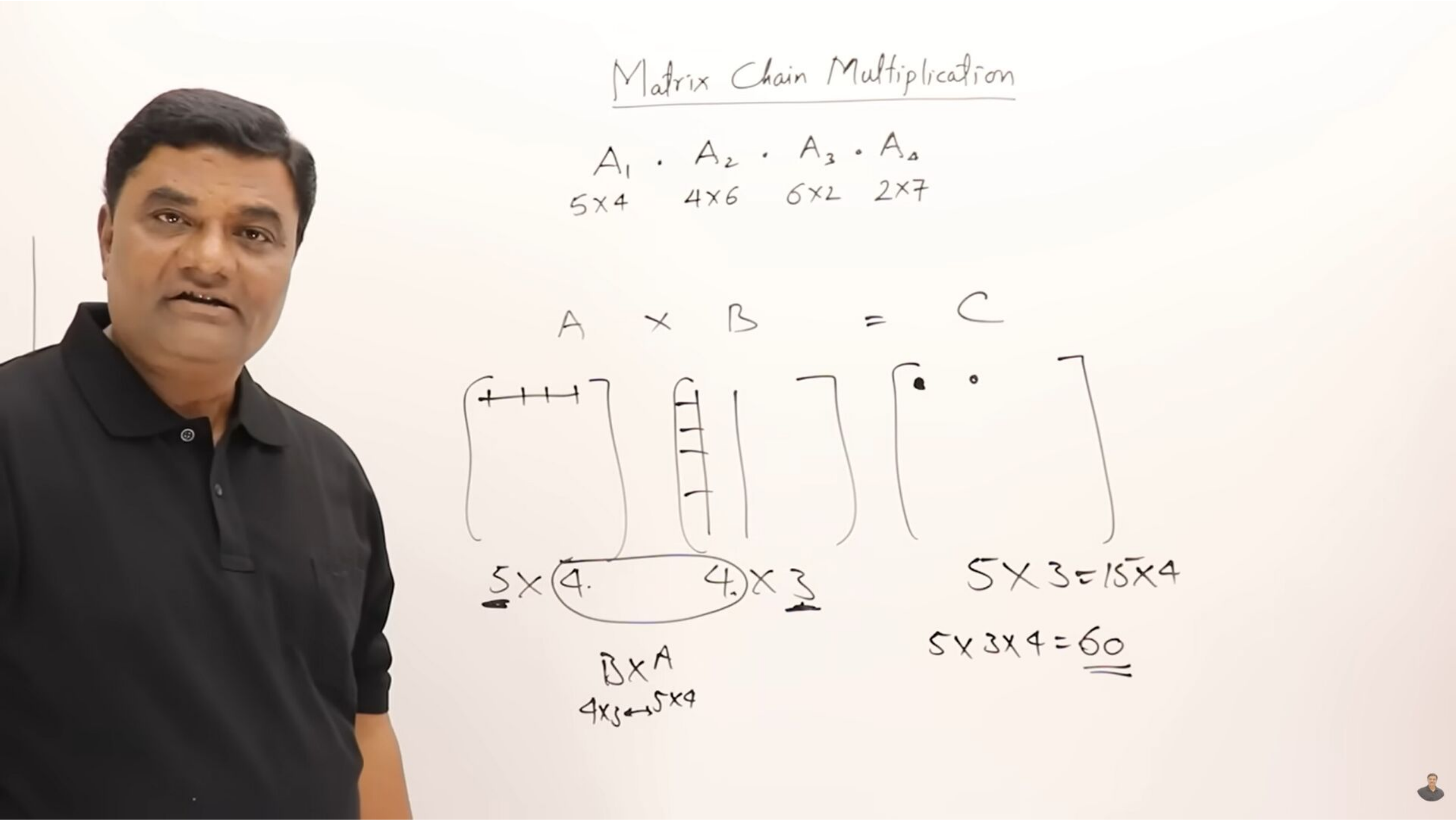
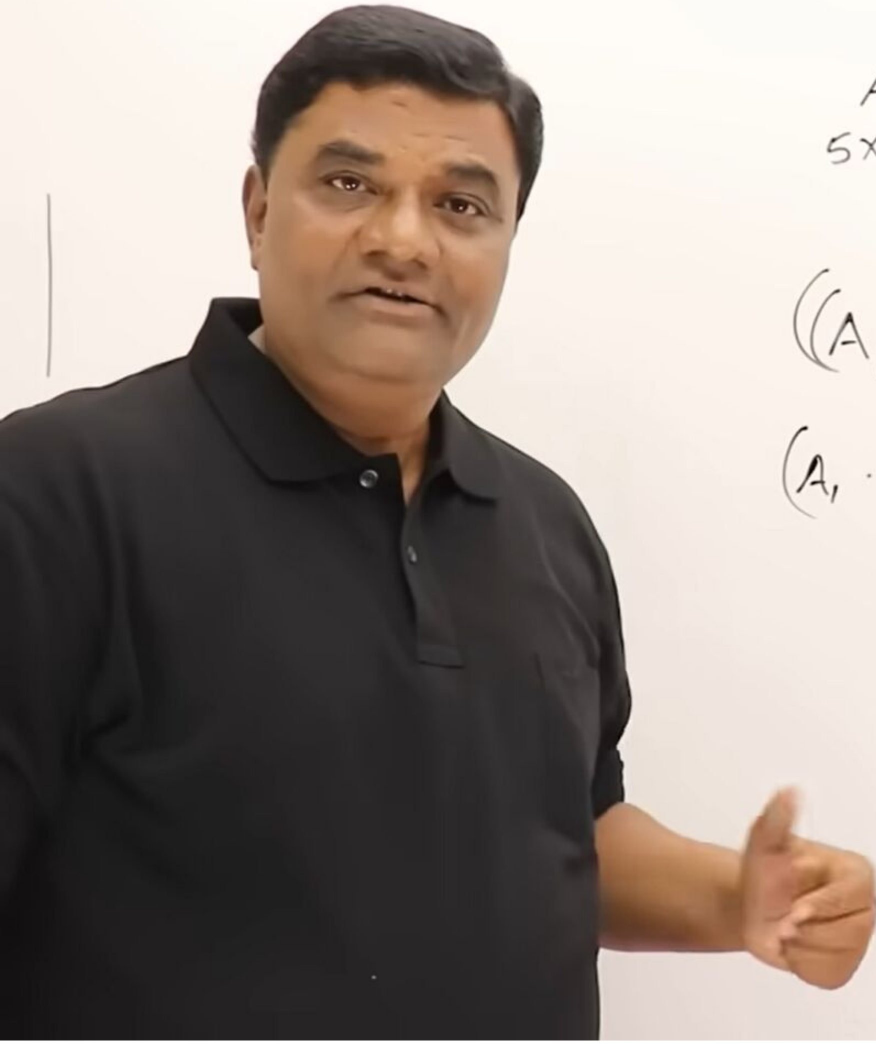

Matrix Chain Multiplication
A1 · A2 · A3 · A4
5×4 4×6 6×2 2×7
A × B = C
5×4. 4.×3
5×3=15×4
5×3×4=60
B×A
4×3↔5×4

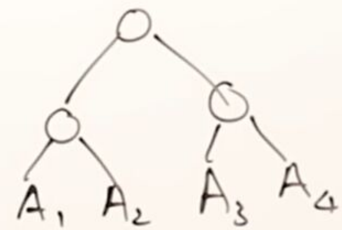
Matrix Chain Multiplication

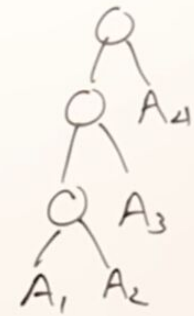


$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$
$$5\times4 \quad 4\times6 \quad 6\times2 \quad 2\times7$$

$$((A_1 \cdot A_2) \cdot A_3) \cdot A_4$$

$$(A_1 \cdot A_2) \cdot (A_3 \cdot A_4)$$





$$T(n) = \frac{{}^{2n}C_n}{n+1}$$

$$T(3) = 5$$

Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$
$$5\times4 \quad 4\times6 \quad 6\times2 \quad 2\times7$$

m	1	2	3	4
1	0	120	88	
2		0	48	104
3			0	84
4				0

$m[2,4] = 104$

$A_2 \cdot (A_3 \cdot A_4)$

$4\times6 \quad 6\times2 \quad 2\times7$

$m[2,2] + m[3,4] + 4\times6\times7$

$0 + 84 + 168$

252

$(A_2 \cdot A_3) \cdot A_4$

$4\times6 \quad 6\times2 \quad 2\times7$

$m[2,3] + m[4,4] + 4\times2\times7$

$48 + 0 + 56$

104

S	1	2	3	4
1		1	1	
2			2	3
3				3
4				

Matrix Chain Multiplication

$$A_1 \cdot A_2 \cdot A_3 \cdot A_4$$

$$\underset{d_0}{5} \times \underset{d_1}{4} \quad 4 \times 6 \quad \underset{d_2}{6} \times 2 \quad \underset{d_3}{2} \times \underset{d_4}{7}$$

$$m[1,4] = \min \left\{ \begin{array}{l} \underset{0 \quad\quad 104 \quad\quad +140}{m[1,1] + m[2,4] + 5 \times 4 \times 7}, \quad \underset{120 \quad + \quad 84 + 210}{m[1,2] + m[3,4] + 5 \times 6 \times 7} \\ \underset{88 \; + \quad 0 \; + 70}{m[1,3] + m[4,4] + 5 \times 2 \times 7} \end{array} \right\}$$

$$m[i,j] = \min \left\{ m[i,k] + m[k+1,j] + d_{i-1} * d_k * d_j \right\}$$

m	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

s	1	2	3	4
1		1	1	3
2			2	3
3				3
4				

Matrix Chain Multiplication

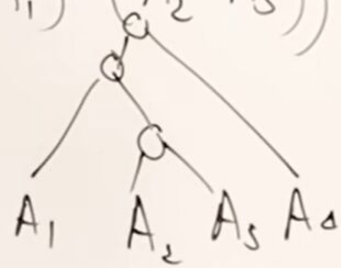
$A_1 \cdot A_2 \cdot A_3 \cdot A_4$

$5 \times 4 \quad 4 \times 6 \quad 6 \times 2 \quad 2 \times 7$

$d_0 \quad d_1 \quad d_2 \quad d_3 \quad d_4$

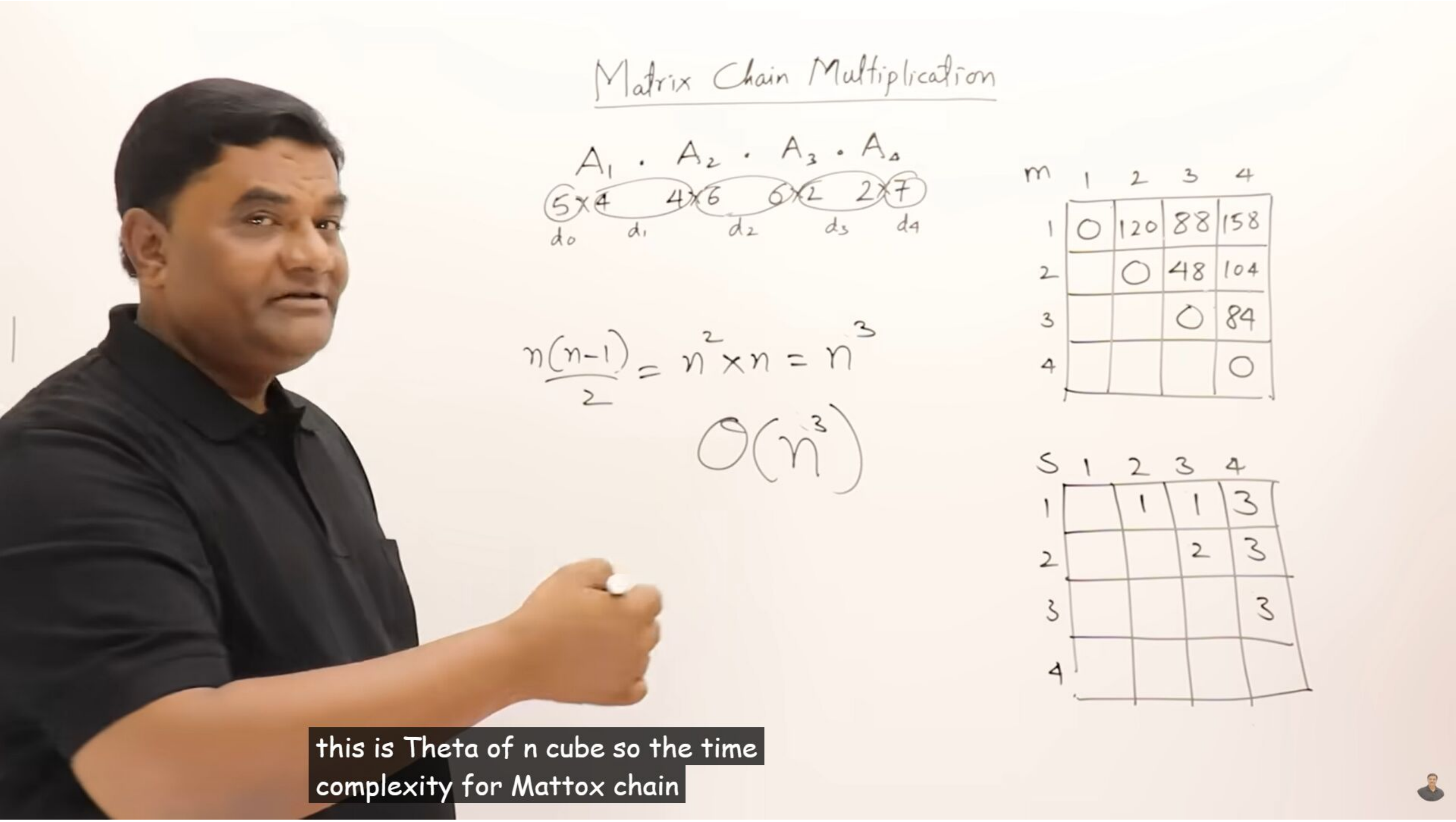
$(A_1 \cdot A_2 \cdot A_3) \cdot (A_4)$

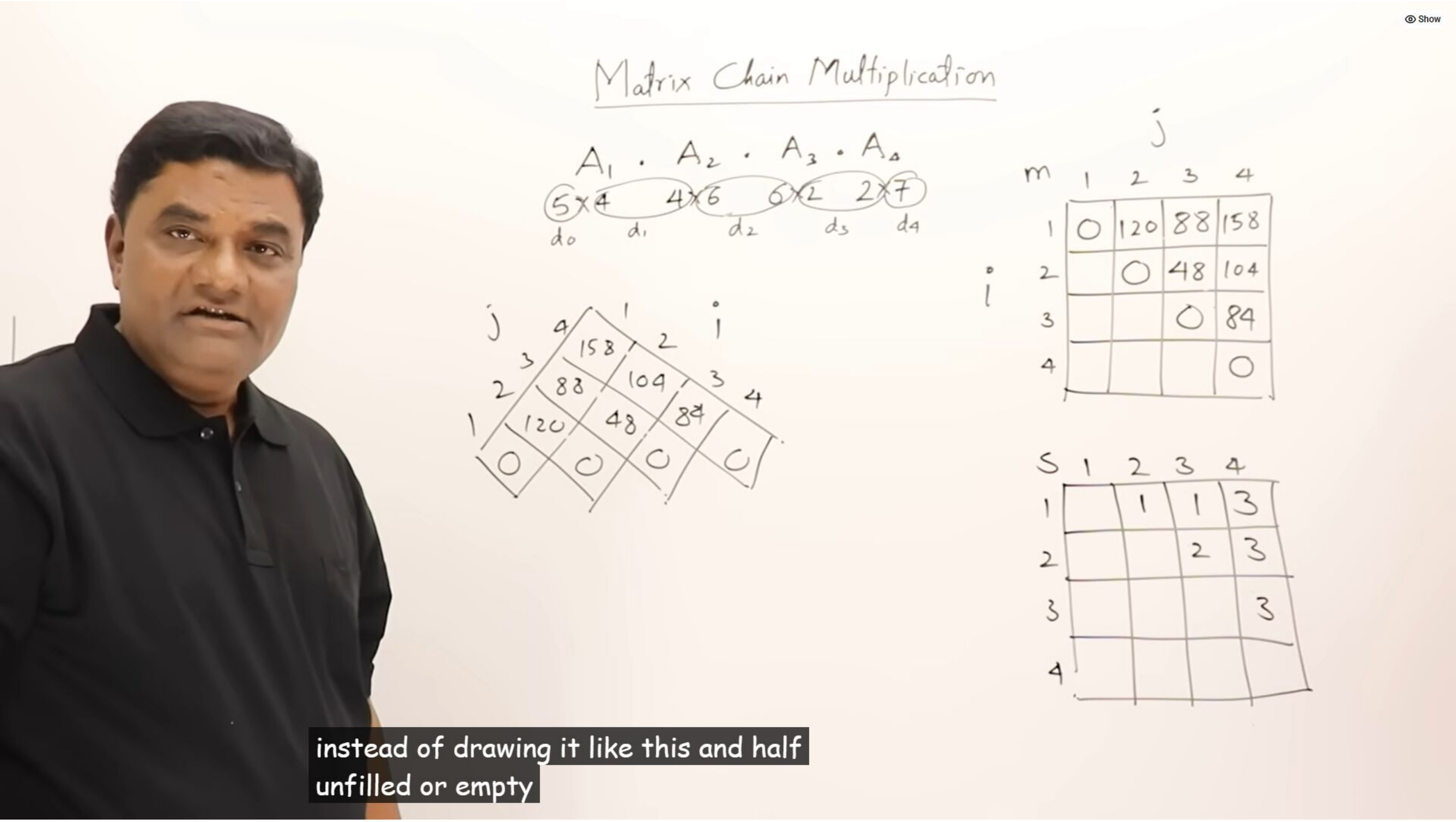
$((A_1) \cdot (A_2 \cdot A_3)) \cdot (A_4)$



m	1	2	3	4
1	0	120	88	158
2		0	48	104
3			0	84
4				0

S	1	2	3	4
1		1	1	3
2			2	3
3				3
4				


Matrix Chain Multiplication
A1 · A2 · A3 · A4
5×4 4×6 6×2 2×7
d0 d1 d2 d3 d4
n(n−1)/2 = n² × n = n³
O(n³)
m 1 2 3 4
1 0 120 88 158
2 0 48 104
3 0 84
4 0
s 1 2 3 4
1 1 1 3
2 2 3
3 3
4
this is Theta of n cube so the time
complexity for Mattox chain


Show
Matrix Chain Multiplication
A1 · A2 · A3 · A4
5x4 4x6 6x2 2x7
d0 d1 d2 d3 d4
m 1 2 3 4
1 0 120 88 158
2 0 48 104
3 0 84
4 0
s 1 2 3 4
1 1 1 3
2 2 3
3 3
4
instead of drawing it like this and half
unfilled or empty